vibrational patterns match the particle properties in Tables 12.1 and 12.2, I think that would convince even the diehard skeptics of string theory's veracity, whether or not anyone had directly seen the extended structure of a string itself. And beyond establishing itself as the long-sought unified theory, with such a match between theory and experimental data, string theory would provide the first fundamental explanation for why the universe is the way it is.

So how does string theory fare on this critical test?

Too Many Vibrations

Well, at first blush, string theory fails. For starters, there are an infinite number of different string vibrational patterns, with the first few of an endless series schematically illustrated in Figure 12.4. Yet Tables 12.1 and 12.2 contain only a finite list of particles, and so from the get-go we appear to have a vast mismatch between string theory and the real world. What's more, when we analyze mathematically the possible energies—and hence masses—of these vibrational patterns, we come upon another significant mismatch between theory and observation. The masses of the permissible string vibrational patterns bear no resemblance to the experimentally measured particle masses recorded in Tables 12.1 and 12.2. It's not hard to see why.

Since the early days of string theory, researchers have realized that the stiffness of a string is inversely proportional to its length (its length squared, to be more precise): while long strings are easy to bend, the shorter the string the more rigid it becomes. In 1974, when Schwarz and Scherk proposed decreasing the size of strings so that they'd embody a gravitational force of the right strength, they therefore also proposed increasing the tension of the strings—all the way, it turns out, to about a thousand trillion trillion trillion (10^{39}) tons, about 100000000000000000000000000000000000000000 (10^{41}) times the tension on an average piano string. Now, if you imagine bending a tiny, extremely stiff string into one of the increasingly elaborate patterns in Figure 12.4, you'll realize that the more peaks and troughs there are, the more energy you'll have to exert. Conversely, once a string is vibrating